

LeetCode 787: Cheapest Flights Within K Stops

Algorithm (Bellman-Ford Approach)

Input: n (cities), flights (u,v,price), src, dst, k

Steps:

1. Initialize distance array dist with infinity and set $\text{dist}[\text{src}] = 0$.
2. Repeat k+1 times:
 - Create temp copy of dist
 - For each flight (u,v,price): if $\text{dist}[u] \neq \text{infinity}$, update $\text{temp}[v] = \min(\text{temp}[v], \text{dist}[u] + \text{price})$
 - Update $\text{dist} = \text{temp}$
3. If $\text{dist}[\text{dst}]$ is infinity, return -1 else return $\text{dist}[\text{dst}]$

Key Logic: Each iteration allows one more edge. After k+1 iterations, paths with at most k stops are considered.

Complexity:

Time: $O(k * E)$

Space: $O(n)$

Summary: Use Bellman-Ford with k+1 relaxations to handle stop constraint.